

backend/core/kv_cache.py Documentation (Roman Urdu)

Yeh file application ke **KV-Cache (Key-Value Cache)** ki state ko track karti hai.

File Ka Maqsad (Purpose)

Asal KV cache Ollama ke ander C++ engine m hota ha, magr backend ko kaisay pta lage ga ke purane message ki history abi b RAM / VRAM mein hai ya remove kr dio gae ha? Yeh script simply ek "Tracker" (timer) k torr pr kam krti ha. Agar user kisi chat me 5 min (300 seconds TTL) baad msg bhejey ga, tou backend fresh request bhejey ga kio k Ollama usko bhool chuka ha. Lkn agar aglay hi sec repy kre ga tou backend "Continuation" (fast resume) kre gha base prompt skip kr k!

Code Blocks aur Classes ki Tafseel

1. class CacheEntry

- **Kya karta hai?:** Eik asaan data class (Metadata) jo specific conversation ki id, tokens, last zikar kab hua tha (`last_accessed`) record rakhti ha.

2. class KVCacheManager

- **Kya karta hai?:** Sari in-memory active chats ki states Dictionary `_cache[conversation_id]` mein hold karti hai.

Tafseel:

- `save_cache_state()`: Chat shuru hone pe entry create karti hai.
- `get_cache_state()`: Chat check karti hai k "ttl_seconds" (jaise 300 sec) poory ho gaye hein tou usy automatically expire/delete maaar kar `None` return krti hai.
- `_evict_stale()`: Memory management ka function hai jo purani entries clean krta hai k list (`_max_entries=20`) se barrh na jaye.